

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE-AI/2024	Date:	

Code of Conduct

I B.VANI(192472348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.VANI

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

DSA0306-NATURAL LANGUAGE PROCESSING

CO4 AT 3

B.VANI(192472348)

Q1. CONSTITUENCY PARSING VS DEPENDENCY PARSING

Introduction

Constituency parsing represents a sentence as a hierarchy of phrases, while dependency parsing represents direct word-to-word grammatical relationships. Both approaches are useful for understanding conversational text, but they focus on different aspects of syntax.

Constituency Parsing

Constituency parsing divides a sentence into nested constituents such as noun phrases (NP) and verb phrases (VP).

Example:

The student reads the book.

S

├── NP

| └── The

| └── student

└── VP

├── reads

└── NP

├── the

└── book

It is useful for identifying hierarchical phrase structure and understanding how groups of words form larger grammatical units.

Dependency Parsing

Dependency parsing represents relationships directly between words using a head-dependent structure.

Example:

reads

|— student

|— book

|— The

|— the

A simplified relationship representation is:

student → reads

reads → book

The → student

the → book

Here, "reads" is the main head, "student" is its subject, and "book" is its object.

Comparison

Feature	Constituency Parsing	Dependency Parsing
Representation	Phrase hierarchy	Word-to-word relationships
Main Focus	Constituents and phrases	Grammatical dependencies
Structure	Tree of nested phrases	Head-dependent tree
Long-Distance Relations	Can represent them indirectly	Represents direct dependencies clearly
Semantic Relation Extraction	Moderate	Strong
Conversational Queries	Useful for phrase structure	Very useful for direct relationships

Suitability for Conversational Text

Dependency parsing is more suitable for extracting meaningful relationships from conversational text because it directly identifies relationships such as:

User → books → flight

Customer → buys → ticket

Person → wants → hotel

For long-distance dependencies, the parser can connect words that are separated by several other words.

Constituency parsing is still useful when the application requires detailed phrase hierarchy or syntactic structure.

Conclusion

Dependency parsing is more suitable for conversational semantic relationship extraction because it provides direct word-to-word relationships. Constituency parsing is preferable when detailed hierarchical phrase structure is required. A practical conversational NLP system can use both representations depending on the task.

Q2. TOP-DOWN PARSING VS EARLEY PARSING

Introduction

An interactive NLP system must process incomplete and ambiguous queries such as:

best hotels near...

book a flight from Chennai to...

Top-down parsing and Earley parsing behave differently when input is incomplete or ambiguous.

Top-Down Parsing

Top-down parsing starts from the grammar's start symbol and predicts possible structures.

For example:

$S \rightarrow NP VP$

$VP \rightarrow Verb NP$

The parser begins with S and attempts to construct the sentence structure.

Problems with incomplete input:

best hotels near...

The parser may expect additional constituents and may repeatedly explore grammar alternatives.

Problems with ambiguity:
Multiple possible grammar rules can cause backtracking.

Earley Parsing

Earley parsing is a chart-based parsing algorithm that maintains partial parsing states. Its major operations are:

Prediction

Scanning

Completion

For:

book a flight from Chennai to...

Earley parsing can preserve the partial structure and continue when more input arrives.

Comparison

Feature	Top-Down Parsing	Earley Parsing
Starting Point	Start symbol	Maintains partial states
Incomplete Input	Less suitable	Highly suitable
Ambiguity	May require backtracking	Can maintain alternatives
Repeated Work	Higher	Reduced using chart
Partial Parsing	Limited	Strong
Interactive Applications	Less suitable	Highly suitable
Complex Queries	Can become inefficient	More robust

Why Earley is Better

Earley parsing can process partially typed queries without requiring the complete sentence. It maintains multiple possible structures and avoids restarting the entire parsing process for every new word.

For example:

book a flight from Chennai to...

The system can keep the existing parse and wait for the destination instead of discarding the partial structure.

Conclusion

Earley parsing is more appropriate for interactive NLP applications because it handles incomplete and ambiguous input effectively, maintains partial structures, and reduces unnecessary backtracking.

Q3. CFG VS PCFG VS NEURAL CONSTITUENCY PARSERS

Introduction

A grammar-checking system must resolve structurally ambiguous sentences. CFG, PCFG, and neural constituency parsers represent and resolve ambiguity differently.

CFG

A Context-Free Grammar defines valid syntactic structures using production rules.
Example:

$S \rightarrow NP VP$

$VP \rightarrow V NP$

$NP \rightarrow Det N$

For an ambiguous sentence, CFG may generate multiple valid parse trees.

Advantage:

It is highly interpretable because every parse can be traced to grammar rules.

Limitation:

It cannot naturally determine which of multiple valid parses is more likely.

PCFG

A Probabilistic Context-Free Grammar assigns probabilities to grammar rules.

Example:

$S \rightarrow NP VP \quad 0.9$

$VP \rightarrow V NP \quad 0.7$

$NP \rightarrow Det N \quad 0.8$

When multiple parse trees exist, their probabilities can be compared and the most probable parse can be selected.

Advantage:

Better ambiguity resolution than CFG.

Limitation:

Performance depends on the quality of probabilities and training data.

Neural Constituency Parser

Neural constituency parsers use machine-learning models to learn syntactic patterns from large datasets. They can use contextual information to select likely parse structures.

Advantages:

1. Strong practical accuracy.
2. Better handling of complex and ambiguous sentences.
3. Can learn patterns that are difficult to encode manually.

Limitations:

4. Less interpretable than CFG.
5. Requires substantial training data and computational resources.
6. Internal decision-making is more difficult to explain.

Comparison

Feature	CFG	PCFG	Neural Parser
Ambiguity Representation	Multiple parses	Multiple parses with probabilities	Multiple candidates ranked by learned model
Ambiguity Resolution	Poor	Good	Very Good
Interpretability	Very High	High	Lower
Training Data	Not essential	Usually required	Usually large
Practical Accuracy	Moderate	Better	Generally highest
Computational Cost	Low/Moderate	Moderate	Higher

Best Balance

PCFG provides the best balance between interpretability and practical accuracy. It retains the understandable grammar-rule structure of CFG while adding probabilities to choose between competing interpretations.

Neural parsers may provide higher practical accuracy, but they are less interpretable.

Conclusion

CFG is highly interpretable but weak at resolving ambiguity. PCFG improves ambiguity resolution through probabilities while remaining understandable. Neural constituency parsers generally provide stronger practical accuracy but sacrifice interpretability. Therefore, PCFG provides the best balance for a grammar-checking system where both accuracy and explanation are important.

Q4. FEATURE STRUCTURES VS TRADITIONAL CFG

Introduction

A multilingual NLP system must represent grammatical information such as number, gender, tense, and person. Traditional CFG rules mainly represent structural patterns, while feature structures attach grammatical properties to linguistic constituents.

Traditional CFG

Traditional CFG may contain rules such as:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

However, these rules alone do not directly enforce detailed grammatical agreement. For example:

The boy plays.

The boys play.

The grammar needs additional mechanisms to distinguish singular and plural combinations.

Feature Structures

Feature structures represent linguistic attributes using key-value pairs.

Example:

NP:

Number = Singular

Person = Third

Gender = Masculine

Verb:

VP:

Number = Singular

Person = Third

Tense = Present

The grammar can require compatible features between constituents.

Example:

NP[Number=Singular] + VP[Number=Singular]

How Feature Structures Extend CFG

Feature structures allow grammatical rules to carry additional information.

Example:

$S \rightarrow NP VP$

NP[Number=N] VP[Number=N]

The variable N ensures that the number of the subject and verb matches.

Similarly:

Person

Gender

Number

Tense

Case

can be represented as features.

Importance in Multilingual NLP

Different languages may have complex agreement systems involving:

1. Number
2. Gender
3. Person
4. Tense
5. Case
6. Grammatical agreement

Feature structures can represent these properties explicitly and ensure

consistency during parsing.

Comparison

Feature	Traditional CFG	Feature Structures
Phrase Structure	Yes	Yes
Number Agreement	Limited	Strong
Gender	Limited	Strong
Person	Limited	Strong
Tense	Limited	Strong
Constraint Enforcement	Weak	Strong
Multilingual Suitability	Moderate	High

Conclusion

Feature structures significantly extend basic grammar representations by attaching linguistic attributes to constituents. They are highly useful for multilingual NLP because they enforce grammatical constraints such as number, gender, person, and tense, improving parsing accuracy and linguistic consistency.

Q5. SUBCATEGORIZATION FRAMES IN VOICE ASSISTANTS

Introduction

Subcategorization frames specify the types and number of arguments that a verb requires. They help a voice assistant determine whether a sentence has a valid syntactic and semantic structure.

1. "Give the book to John."

The verb "give" requires multiple arguments.
Subcategorization frame:

give → Subject + Object + Recipient

Semantic representation:

Give(Agent, Book, John)

Here:

Agent = Person performing the action

Object = Book

Recipient = John

The sentence is valid because all required arguments are present.

2. "Sleep."

The verb "sleep" does not require a direct object.

Subcategorization frame:

sleep → Subject

Example:

The child sleeps.

Adding an unnecessary direct object would create an incorrect structure such as:

The child sleeps the book.

3. "Put the file on the desk."

The verb "put" requires a direct object and a location.

Subcategorization frame:

put → Subject + Object + Location

Semantic representation:

Put(Agent, File, Desk)

All required arguments are present.

Comparison

Verb	Required Arguments				Example
Give	Subject	+	Object	+	Give the book to John
	Recipient				
Sleep	Subject				Sleep
p					

Put Subject + Object + Put the file on the
 Location desk

Importance in Syntactic Analysis

Subcategorization frames help the system:

1. Determine whether a sentence is grammatically valid.
2. Identify required arguments.
3. Detect missing arguments.
4. Distinguish optional and required phrases.
5. Construct more accurate parse structures.

Importance in Semantic Analysis

Subcategorization also helps identify semantic roles.

For "give":

Agent → gives

Object → thing given

Recipient → receiver

For "put":

Agent → moves object

Object → file

Location → desk

This improves semantic interpretation of voice commands.

Voice Assistant Example

User:

Give the book to John.

Structured representation:

Action = Give

Object = Book

Recipient = John

User:

Put the file on the desk.

Structured representation:

Action = Put

Object = File

Location = Desk

The assistant can therefore convert natural-language commands into structured actions for downstream processing.

Conclusion

Subcategorization frames are important because they specify the argument requirements of verbs. They improve both syntactic parsing and semantic interpretation by identifying valid sentence structures, required participants, and their semantic roles. They are particularly useful in voice assistants where accurate command interpretation is essential.